PRODUCT BLUEPRINT · PLAIN-LANGUAGE EDITION · REVISION 3 (FINAL)

PG Management System

How the whole system works — who does what, and how the day-to-day runs. Written so anyone can follow it, not just developers. **Revision 3** is the authoritative version: phone-only sign-in, three roles plus a platform operator, and the public PG marketplace fully designed.

> **What's new in revision 3**
> Sign-in is now **phone number + one-time code only** — no passwords, no invite codes anywhere. The idea of "staff" is gone; there are three roles in the app, driven by privileges. Managers see only their own branch, by design. And the **public PG marketplace** is fully specified: anonymous browsing, live vacancy, a lead inbox for managers, owner security, and platform-controlled listings that you get paid for.

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1 · The big picture

Hamsa Vilas runs paying-guest accommodation across several branches. Today that usually means a paper register, spreadsheets, and WhatsApp groups. This system brings it all into one place: rooms and beds, residents and their rent, cash payments and receipts, complaints, and move-outs — and it puts the PG on a public map where new residents can find it.

- **One system, three ways to open it** — a website, an Android app, and an iPhone app. They all share the same information behind the scenes, so nothing goes out of sync. *(The Android app is complete today; web and iPhone follow.)*
- **Everyone sees only what they should** — a resident sees their own stay, a branch manager runs their branch, the owner oversees the whole business, and the public sees only what's been listed.
- **Cash-first, made simple** — rent is collected in cash; the manager records it and the resident instantly gets a downloadable receipt.
- **One key opens every door: a phone number.** The same phone signs in a resident, invites them, and follows them from "just browsing" to "living here" — one identity, no passwords.

In plain words: a single shared notebook for the whole PG business — it can't be lost, it remembers who wrote what, it shows each person only their own page, and it has a shop window open to the street.

2 · The words we use

WORD	WHAT IT MEANS (WITH AN EXAMPLE)
Organization	The whole PG business that owns all the branches. Example: Hamsa Vilas.
Branch	A single PG location. Example: Hamsa Vilas – Koramangala.
Owner	The boss of one PG business. Can do anything across every branch — and approves the managers' changes.
Manager	Runs one branch and handles all the daily work there.
Tenant	A resident — a paying guest who lives in a branch.
Prospect	A person searching for a PG who has enquired but hasn't moved in yet. A light identity — just a phone number — that becomes a Tenant on the same number when they join.
Bed	What a resident is actually assigned to, and the unit all pricing is quoted against. Beds sit inside rooms and carry a short label ("A", "B").
Listing	A branch's public face on the marketplace: title, address, map pin, description, amenities, and live vacancy.
Platform Operator	You — the business running this app. Sits above every PG. Approves and publishes listings, and gets paid for them.

One word to watch: "tenant" means a resident (the everyday meaning), not the company.

3 · How it's all connected

The business sits at the top, branches sit under it, rooms and beds sit inside each branch, and residents live in the beds. The owner sees everything within their business; each branch has its own manager. Above all the businesses sits you, the platform.

> **PLATFORM OPERATOR** (you — approves & publishes listings, bills) → **OWNER** (full control of one PG, approves changes) → **ORGANISATION**: Hamsa Vilas → **BRANCH** (e.g. Koramangala) + its Manager → **Rooms & Beds** → **Tenants**

In one line: you run the marketplace; each PG owner runs their branches; each branch has a manager, rooms and beds; residents live in the beds. A single-PG owner is just an organisation with one branch.

Where the organisation itself comes from

A new PG business is set up in a single privileged step that creates the organisation and its first owner. This is a one-time back-office action, not self-service — nobody can spin up a PG business from inside the app.

Which branch am I working in?

An owner works across many branches, so the app asks once which branch they're working in, then applies that choice everywhere — occupancy, payments and the dashboard all follow the same scope. This matters most on payments: recording cash against the wrong branch is painful to unwind. A manager never sees this choice — they're pinned to their own branch.

4 · The Tenant (resident)

A resident's app is all about their own stay — nothing else.

What a tenant CAN do

- **Sign in with their phone number and a one-time code.** They enter their number, get a code by SMS, type it in — done. No password to remember, nothing to reset. The manager registered that phone when they joined, so their number simply works.
- **Prove who they are once, at first login** — DigiLocker, phone, and an emergency contact. The app holds them here until it passes; nothing else opens first.
- **See exactly how much rent they owe** — with the full picture: rent, deposit held, total paid, accrued so far, and outstanding balance.
- **See and download all their receipts, any time**, as PDF files, in their own Receipts tab.
- **Raise a complaint and follow its progress** until it's fixed.
- **Tell the PG they'll be away** for a few days (a safety feature — see workflow note).
- **Read a Help & FAQ page** covering when rent shows as due, how deposits come back, and who can see their details.

What a tenant CANNOT do

- Sign up on their own — a manager registers them first.
- See other residents' details or complaints.
- Pay online — payments are cash for now.
- Change their own name, phone or emergency contact — the manager updates these straight away.

Why it works this way: residents can't invite themselves, and every new resident is verified once, so the PG always knows who is actually living there.

5 · The Manager

The manager is the engine of a branch — they do the day-to-day work for their one branch, and nothing outside it.

What a manager CAN do

- Run their assigned branch and everything in it.
- **Register new residents by phone number.** They enter the resident's Aadhaar-exact name and phone; that phone can now sign in. No code to pass on, no link step — the number *is* the invitation.
- **Record cash payments** — the resident instantly gets the receipt. Amounts are typed in freely, so **part-payments** work and the balance carries forward.
- **Set each resident's rent and deposit** when the resident is added.
- **Create rooms and beds, and assign a resident to a bed** — or free it again — by tapping the bed. Occupancy is

shown per room.
- **Handle complaints from start to finish**, along the fixed path in workflow 3.
- **Send notices to their branch** ("rent due on the 5th", "water off tomorrow").
- **Handle move-outs and settle deposits** — last day, notice given, and any itemised deductions.
- **See their branch's money** — collected, deposits held, accrued, outstanding, and active residents.
- **Work the enquiry inbox** — every lead from the public marketplace for their branch lands here (see section 11).
- **Prepare a public listing** for their branch and send it for publishing — but not publish it themselves (see section 7).

The one rule a manager lives by — "maker-checker"

- **Adding brand-new things is instant**, no permission needed.
- **But changing or deleting anything needs the owner's approval first.**
- The manager is told immediately which happened — the change either applies on the spot, or the app confirms it's been sent for approval. Nothing fails silently. This covers branches, rooms and residents.

What is "maker-checker"? Two people sign off on a risky action. The manager makes the change; the owner checks and approves it. This stops important records (like a resident's dues) being altered or wiped by one person alone.

Branch isolation is deliberate: a manager sees their own branch and nothing about any other — not its residents, not its rooms, not its availability. This is by design, not a limitation.

6 · The Owner

The owner sits above every branch of their business and has the final say within it.

What an owner CAN do

- **Do everything, across every branch** — add, change, delete — with no approval needed.
- **Approve or reject the managers' changes and deletions**, from one pending list showing the record, the operation, and who asked.
- **Add people to the business** — a manager (tied to the branch they'll run), another owner, or a resident. The rule is simple: **you can only create roles at or below your own**. A manager, by that rule, can add residents only. There is no separate "staff" process — it's just adding a person and picking their role, and the app shows each person what their role allows.
- **See all the money:** each branch on its own, and the whole business combined.
- **See every complaint** — including private business-level ones managers can't see.
- **See enquiry leads across all branches** (managers work their own branch's leads; the owner sees the whole picture).
- **Set the receipt branding** — display name, address, footer, accent colour — on every receipt. Owner-only.
- **Read the audit log** — who did what, to which record, and when. Owner-only.
- **Send notices to the whole organisation**, not just one branch.

7 · The Platform Operator (you)

This is the actor above every PG business — the company running this app. It exists because the public marketplace is *your* product, and listings are how it earns.

What the platform operator CAN do

- **Approve and publish listings.** An owner or manager can *prepare* a listing — write the description, set amenities, drop the map pin — but it stays a draft until you approve it. Nothing reaches the public without you.
- **Bill for listings.** Because you control what appears, you control what's charged for. This is the monetisation point of the whole marketplace.
- **Take a listing down** or ask for changes before it goes live.

> **Why publishing sits with you, not the PG**
>
> If PGs published themselves for free, the marketplace would be a feature they use. Because **you** are the gate, it's a business you run: a PG pays to appear, appear higher, or be featured, and your approval is what they're paying for. It also protects the brand — no PG can push an inaccurate address, a bad photo, or a stale claim in front of the public without a check.
>
> **Billing model to decide (not needed to launch):** a flat monthly listing fee (simplest to sell first), pay-per-lead (the enquiry inbox already counts leads cleanly — the most defensible), or featured-placement. You can start flat and add pay-per-lead once lead volume is real.

8 · How the main workflows run

1. Getting registered & verified

Manager registers the person by **Aadhaar-exact name + phone number** → that phone is now allowed to sign in → the resident opens the app, enters the same phone, gets a one-time code by SMS, types it in → first login lands on the verification gate: DigiLocker + phone + emergency contact → account active

In one line: the manager adds a resident by phone, the resident signs in with that phone and an SMS code, and proves their identity once — no password, no invite code, nothing to pass on.

When verification doesn't pass: the check returns a reason the resident can read; they stay on the gate and can retry or sign out. The step also carries an optional country, leaving room for a non-Aadhaar path for foreign residents later. *Still to decide: how many retries before a manager steps in, and the manual override.*

2. Paying rent (cash → instant receipt)

Resident pays cash at the branch → Manager confirms the amount and marks "payment received" → Receipt created instantly with its own number → Resident views & downloads it as a PDF (dues update automatically)

In one line: cash is handed over, the manager records it, and the resident immediately has a receipt and an updated balance. Part-payments are allowed; the rest stays outstanding.

3. Raising & resolving a complaint

Open → Assigned → In progress → Resolved → Closed

In one line: a complaint moves through clear stages so the resident always knows where it stands — and can bounce back to "in progress" if it wasn't really fixed.

FROM	CAN MOVE TO
Open	Assigned, or straight to In progress
Assigned	In progress
In progress	Resolved
Resolved	Closed — or back to In progress if it wasn't really fixed
Closed	Nowhere. Closed is final.

A complaint can be marked **business-level** when raised — those are visible to the owner only, never to managers.

4. Moving out & getting the deposit back

Resident gives notice → Staff record the last day and notice given (default 30 days) → Enough notice? Yes → refund eligible / No → deposit forfeited → Settlement: rent to last day + dues → Deposit returned minus fair, itemised deductions (cash) → Marked moved-out — bed becomes available

In one line: give proper notice and you get your deposit back (minus fair deductions); skip the notice and it may be forfeited.

The settlement returns the full breakdown — rent to the last day, total paid, outstanding, deposit held, deductions total, final refund, whether forfeited, and a short note — so the conversation at the desk is the same every time.

5. Getting found & enquiring (the marketplace)

Seeker browses listings anonymously → finds a PG, sees address, map, amenities, live vacancy → taps "Enquire" → enters phone + one-time code (their only step) → becomes a **Prospect** → the lead lands in that branch's **Manager enquiry inbox** → manager follows up → if they move in, the manager registers them on the **same phone** and the Prospect becomes a Tenant

In one line: anyone can browse without signing in; the moment they enquire they give a phone and a code, and that same number carries them all the way to becoming a resident.

6. Maker-checker (how a manager's change gets approved)

Manager edits or deletes something → Change waits in a "pending approval" list → Owner approves? Yes → change goes live / No → dropped, record unchanged

In one line: a manager can add things freely, but any edit or delete waits for the owner's yes.

9 · Money & billing

Three ways to stay

Every resident is put on one of three plans when they join. Pricing is always per bed, so shared rooms split naturally.

PLAN	WHO IT'S FOR	HOW IT'S BILLED
Monthly	Regular long-stay residents (the usual)	Per-bed monthly rent, billed each month; part-months charged by the day
Weekly	Shorter stays	Per-bed weekly rate, billed each week
Daily	Very short stays / guests	Per-bed daily rate, usually paid upfront

When rent shows up as due

A month's dues appear from the **30th of the previous month** — so August's rent shows from 30 July. Residents are told the same thing in the app, because it's the most common question, and it means an outstanding figure runs slightly ahead of the calendar.

How amounts are held

Every amount is stored as a whole number of paise and only shown as rupees, with Indian grouping (₹1,25,000). Nothing is ever a decimal that could drift — the ledger adds up to the paisa, every time.

The security deposit

- **At join:** collected and recorded separately from rent — the resident sees "Deposit held: ₹X".
- **At move-out:** returned minus fair, itemised deductions (unpaid rent, damage, cleaning), each with a reason.
- **The catch:** the deposit only comes back with proper notice. No notice can mean forfeiture.

Why the itemised list matters: deposits are the #1 thing residents and PGs argue about. A clear "collected ₹X, deducted ₹Y for these reasons, refunded ₹Z" record settles it before it becomes a fight.

Who sees which money

- **Tenant** — only their own dues and receipts.
- **Manager** — their branch: collected, deposits, accrued, outstanding, active residents.
- **Owner** — every branch, plus the whole-business total.
- **Platform Operator** — listing fees and lead counts, not any PG's internal ledger.

Receipts carry the business's branding

Every receipt is a downloadable PDF that picks up the organisation's display name, address, footer and accent colour from one template the owner controls. Change it once and every receipt after follows.

10 · Rooms, beds & floors

- A **branch** holds **rooms** (each with a number); a **room** holds **beds** (each with a short label). A resident is **assigned to a bed** by tapping it, and freed the same way. A bed shows its occupant when taken.
- Pricing is per bed, which is what makes shared rooms split cleanly.

> **Floors are a naming convention, not a stored record**
> A room's floor is read from the first character of its number: `101` → floor 1, `204` → floor 2, `G01` → ground. Anything that doesn't follow the pattern groups under "Other rooms". **This makes room numbering a business rule** — every branch must adopt the convention, or floor grouping becomes meaningless.

11 · The public PG marketplace

The one part of the system open to the street. It's where new residents find the PG — and, through the Platform Operator, where the app earns.

Browsing is free and needs no account

Anyone can list PGs, search by area or city, ask for the ones **nearest to them** (which orders results by distance), and open a listing's full detail — with **zero sign-in**. A login wall here would lose seekers before they've compared even two PGs, so there isn't one.

Full address is shown up front — on purpose

People choose a PG by how far it is from their office or college, so the **full address and precise map pin are public**. Hiding them would break the main reason people search. No "reveal after enquiry" games.

Vacancy is live, not hand-typed

Because the system already knows every bed's occupancy, vacancy is **derived automatically** — "4 beds available", straight from the real data, shown with the starting price ("3 beds free from ₹8,000"). It can never go stale from someone forgetting to update it, which is the single biggest complaint about every rival's listings. This is a real edge; lead with it.

Acting takes one step: phone + code

The moment a seeker wants to **enquire** (or shortlist, or ask to be told when a bed frees up), they enter a phone number and a one-time code — the same flow the whole app uses. That single step turns an anonymous browser into a **Prospect**: a light identity, no password, no profile form. Capturing the phone here is the point — the phone number *is* the business value of a lead.

One identity, all the way through

Because both enquiring and moving in key on the phone number, there's **one identity across the whole journey**: anonymous browser → Prospect (enquired) → Tenant (moved in). When the person actually joins, the manager registers them on the same phone they enquired with — no second account, nothing to reconcile.

Every lead lands with the branch manager

An enquiry is worthless if no one sees it, so each lead drops into an **enquiry inbox worked by the branch manager** of the PG enquired about — name, phone, which listing, when. Managers field walk-ins already, so leads sit naturally with them. The **owner sees all leads across branches** for the wider picture. This inbox is what makes the marketplace pay for itself; without it, interest evaporates.

Publishing is controlled by the platform

Owners and managers **prepare** a listing; **you publish it** (section 7). This keeps quality high and is where billing sits.

12 · Signing in & account security

One way in, for everyone: phone + one-time code

There are no passwords and no invite codes anywhere in the system. A person is registered by phone — by a manager (residents), by an owner (managers, owners, residents), or by themselves enquiring on the marketplace (prospects) — and from then on they sign in by entering that phone and the code texted to it. This is simpler to use, simpler to support (nothing to reset), and actually **more secure** than a password, because a resident's account is also bound to a one-time DigiLocker identity check.

Extra protection for owners, in language they already know

Owners hold the money and the approval power, so their account gets a second layer. The risk being closed is SIM-swap — someone hijacking the phone number to receive the code.

LAYER	WHO	WHAT IT FEELS LIKE
App PIN `required for owners`	Owners (managers optional)	Set a 4-digit PIN once; enter it to open the app — exactly like a banking or UPI app. A SIM-swap attacker gets the code but not the PIN, so they're still locked out. This is the non-negotiable one.
Biometric unlock `optional`	Anyone whose phone supports it	A thumbprint or face-scan instead of typing the PIN. A convenience on top; the PIN is always the fallback.
Trusted device `automatic`	Everyone	The owner's own phone is remembered; a sign-in from unfamiliar hardware asks for an extra confirmation. Invisible on a normal day.

Deliberately avoided: authenticator apps (confuse non-tech users) and a second SMS code (still SIM-swappable). Managers can be given the PIN too if a business wants it, but it's required only for owners.

13 · Rules that keep it safe

- **Real people only.** Every resident is verified once via DigiLocker, registered with their Aadhaar-exact name — records always match the real person, and the app blocks everything else until it passes.
- **Privacy by default.** Residents can't see each other's complaints; a manager can't see other branches at all; private business-level complaints are hidden from managers.
- **Nothing is truly deleted.** A "deleted" item is only hidden. Once a month, deleted items move to cheap long-term storage, so anything can be recovered. *(This monthly move still needs an owner: a scheduled job and a retention period — one for your CA.)*
- **Every change is on the record.** The system logs who changed what and who approved it; the owner can read that trail in the app.
- **Sessions expire and renew quietly.** Sign-in issues a short-lived pass and a longer renewal pass; the app renews in the background and the person never notices. The renewal pass rotates each time, so a stolen one is near-worthless. Signing out is registered on the server.
- **Traffic is encrypted end to end;** plain connections are refused everywhere except a developer's own machine.
- **Identity documents never leave the server.** The app is only ever told *whether* a person is verified — never their Aadhaar number, DigiLocker reference or emergency contact. Nothing sensitive sits on the phone.

14 · Who-can-do-what, at a glance

ACTION	PLATFORM	OWNER	MANAGER	TENANT	PROSPECT
Browse the public marketplace	Yes	Yes	Yes	—	Yes (no login)
Enquire on a listing	—	—	—	—	Yes (phone + code)
Publish a listing	Yes	Prepare only	Prepare only	No	No
Bill for listings	Yes	No	No	No	No
Work the enquiry inbox	—	All branches	Own branch	No	No
Add new records (rooms, beds, residents…)	—	Yes	Yes	No	No
Edit / delete records	—	Yes	Needs approval	No	No
Approve a manager's change	—	Yes	No	No	No
Add a person	—	Owner / Manager / Tenant	Tenant only	No	No
Record a cash payment	—	Yes	Yes	View own	No
Assign / free a bed	—	Yes	Yes	No	No
See a branch's money	—	All	Own	Own dues	No
See whole-business money	—	Yes	No	No	No
See other branches	—	Yes	No — none	No	No
Handle complaints	—	Incl. business-level	Own branch	Raise own	No
Send notices	—	Org-wide or branch	Own branch	Receive	No
Receipt branding · audit log	—	Yes	No	No	No
Sign-in method	Phone + code + PIN	Phone + code + **PIN**	Phone + code (PIN optional)	Phone + code	Phone + code

15 · Decided vs. still to decide

Locked in

- Roles: Owner, Manager, Tenant in-app (privilege-driven, no separate "staff"); Platform Operator above them for the

marketplace.

- Sign-in is phone + one-time code for everyone. No passwords, no invite codes. Registration by phone is the invitation.
- Owners require an app PIN; biometric optional; trusted-device automatic. Managers' PIN optional.
- Managers are scoped to their own branch only — by design.
- Maker-checker on edits & deletes; creates are free.
- Monthly / weekly / daily plans; per-bed pricing; amounts in paise.
- Cash-only payments, instant PDF receipts, part-payments allowed.
- Move-out notice controls the deposit refund; deductions itemised with reasons.
- Nothing hard-deleted; monthly archive; full audit log readable by the owner.
- Dues appear from the 30th of the previous month; floors derived from room number.
- **Marketplace:** anonymous browsing with full address + live auto-derived vacancy; phone-OTP only to act; one phone identity from browser → prospect → tenant; leads worked by the branch manager (owner sees all); listings published only by the Platform Operator, which is where billing sits.

Still to confirm

- Do daily/weekly guests pay a deposit & do full ID checks?
- Are short stays paid upfront or at checkout?
- DigiLocker: how many retries before a manager steps in, and the manual override (a reason is already shown on failure).
- Can a foreign resident verify without Aadhaar? (The step already carries a country; no alternative path is defined yet.)
- Notices: read-receipts now, scheduling later?
- The monthly-archive job and how long to keep archived records (ask your CA).
- **Marketplace billing model:** flat monthly listing fee, pay-per-lead, or featured placement — start flat, add pay-per-lead once lead volume is real.
- Should rent & deposit be editable after a resident is created (routed through maker-checker)? Recommended — annual rent revisions are routine.

Known to be still-to-build

- "Away days" visible to staff (today saved on the resident's phone only — needs a home on the server before the safety-feature claim is fully true).
- Push notifications (notices are seen only when a resident opens the app).
- Deleting a bed (beds can be created but not removed).
- Web & iPhone clients (Android is complete; the other two follow).
